



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Smart Driver Drowsiness & Safety System

Team ID: DSCPP-III-2026-T234

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026-27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T234
Semester: 3rd
Section: H / E / ML4 / ML3
Domain: C++ & Computer Vision

Team Members

1. *RIDVIK GARG – 2028039*
2. *Prabhav Gulyani – 2027975*
3. *Nandini Gupta – 2028777*
4. *Antriksh Gehlot – 2028648*

Mentor

Mr. Kamal Kumar Gola

Interactions completed: _____

Problem Statement & Motivation

Problem Statement

If a driver gets tired, they can lose attention on the road, especially on long night trips. Many existing student projects only check for closed eyes and just give a basic beep sound, which is not enough.

Motivation

- ▶ *To stop accidents that happen when drivers fall asleep.*
- ▶ *Current small projects don't check distraction and sleepiness together.*
- ▶ *Just an alarm is not enough; we need music control and SMS alerts.*

Target Users / Stakeholders

- ▶ *People driving at night or on long journeys.*
- ▶ *Family members who want to ensure the driver's safety.*
- ▶ *General road safety for everyone.*

Objectives

Project Objectives

1. *Check if the driver's eyes stay closed for too long.*
2. *Find out if the driver is looking away from the road (distraction).*
3. *Give a normal warning first, and a loud alarm if they ignore it.*
4. *Pause or lower the car's music volume during the warning.*
5. *Send a text message to a saved family contact in critical cases.*

Key Objective: *To build a working safety prototype that monitors the driver, warns them, controls music, and saves the events.*

Proposed Solution

Proposed Approach

1. *Input: Camera captures the driver's video feed.*
2. *Processing: OpenCV checks the face and eyes.*
3. *System Logic: C++ code decides if the driver is sleepy.*
4. *Database: SQLite saves the history of warnings.*
5. *Output: Speaker alarm, music control, and family SMS.*

Instead of just sounding an alarm, our system lowers the music so the driver can actually hear the warning.

Key Features

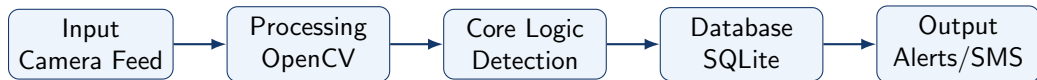
- ▶ *Checks both sleepiness and looking away.*
- ▶ *Automatically stops or lowers the music.*
- ▶ *Emergency message feature for family.*
- ▶ *Shows live status (Good, Warning, Critical).*
- ▶ *Uses Linked List, Queue, and Stack for data.*

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	<i>Linked List (history), Queue (events), Stack (recent alerts)</i>	<i>Event Manager</i>
OOPs with C++	<i>C++ classes, objects, and OpenCV functions</i>	<i>Core Logic</i>
Operating Systems	<i>Threads to run camera and alerts at the same time</i>	<i>Processing Layer</i>
DBMS	<i>SQLite for storing driver details and past records</i>	<i>Database Layer</i>

Requirement: Demonstrate meaningful integration of both subjects in **one** project.

System Architecture / Workflow



Major Components

- ▶ *Camera input module*
- ▶ *Face and eye tracking logic*
- ▶ *Alert and Event handling system*

Data / Control Flow

- ▶ *Camera sends frames to OpenCV.*
- ▶ *Code checks for closed eyes or distraction.*
- ▶ *If unsafe, it triggers alarm, stops music, saves event.*

Technology Stack & Methodology

Technology Stack

- ▶ Programming: *C++*
- ▶ Database: *SQLite*
- ▶ Tools / Framework: *OpenCV*
- ▶ IDE: *VS Code*
- ▶ Version Control: *Git / GitHub*

Development Methodology

1. *Understand requirements and plan modules*
2. *Write code for OpenCV face detection*
3. *Add logic for alerts and music control*
4. *Connect SQLite DB and Data Structures*
5. *Final testing and bug fixing*

We chose C++ and OpenCV because they are fast and good for real-time video processing.

Initial Progress & Team Contribution

Progress Achieved

- ▶ *Decided project requirements and modules*
- ▶ *Designed the basic system architecture*
- ▶ *Planned how to use Queue, Stack, and Linked List*
- ▶ *Created the SQLite database tables*
- ▶ *Tested basic camera input with OpenCV*

Individual Contribution

Member	Role	Contribution
RIDVIK	Team Lead	Core Logic & Architecture
Prabhav	Developer	OpenCV Drowsiness Code
Nandini	DB/Testing	SQLite & Distraction Code
Antriksh	Developer	Music Control & Alerts

Roadmap, Expected Outcomes & References

Roadmap

1. *Phase-I: Project planning and basic design*
2. *Phase-II: Writing OpenCV and Database code*
3. *Phase-III: Joining all parts and final testing*

Expected Outcomes

- ▶ *A working prototype for driver safety*
- ▶ *Working music control and SMS alert feature*
- ▶ *Live display showing Good*

References

1. *OpenCV Documentation Computer Vision Library*
2. *SQLite Documentation Lightweight Database Engine*
3. *C++ Reference Standard C++ Language and Library*
4. *OS concepts related to threads and synchronization*

Thank You
Questions & Suggestions